timeseries data such as temperatures, pressures, flow rates, motor speeds, and valve positions. This data is stored in a data store known as a *process historian* that is designed to handle the large volume of timeseries data generated by industrial processes over time and provide for analysis of past performance and trends.

Given the high-frequency nature of SCADA data (often multiple readings per second), most digital twins import and store this data locally rather than querying SCADA systems or the process historian directly. Direct queries could impact SCADA performance, and the query interfaces often aren't designed for analytical workloads. Instead, organizations typically deploy data collectors that subscribe to SCADA data streams, buffer the data, and forward it to a database that maintains a copy of the operational data known as a *mirror*. A digital twin typically sources data from this mirror rather than directly from the process historian itself.

INDUSTRIAL INTERNET OF THINGS (IIOT)

You may be wondering where the IoT sensors that we have looked at fit into the operational technology landscape. While IoT sensors enable increasingly extensive monitoring of the physical environment, my LoRaWAN flow meter is not part of an OT system. There are some key differences between OT systems and IoT sensors:

- *Connectivity* - OT systems traditionally operate on proprietary networks that are isolated from the internet for security and reliability, while IoT devices normally connect via the public internet.
- *Core function* - OT systems are designed for the direct, real-time management and control of physical systems whereas IoT devices are primarily responsible for data gathering, communication, and non real-time control.

Despite these differences the lines between OT and IoT are beginning to blur with IoT devices increasingly being used to supplement operational decision making in OT systems via the *industrial internet of things (IIoT)*. IIoT devices are connected devices specifically designed for industrial environments and are characterized by their rugged build, interoperability with legacy industrial control systems, and secure design.

4.2.2 Information technology (IT) data sources

While OT systems provide the real-time operational data that drives digital twin monitoring and control, IT systems contribute the business context and enterprise intelligence that transform raw operational data into actionable insights. IT systems often support more flexible access patterns than OT systems, allowing digital twins to choose between in-place access and import based on specific requirements.